1 Linear Regression with Regularization [10 marks]

You will predict `GPA` (Refer Assignment 1 for dataset) using student features. Use a validation set to select the hyperparameters.
Start with a function of the following form:

```
def run_poly_regression(X_train, y_train,
                        X_val, y_val,
                        X_test, y_test,
                        degree=1,
                        regularizer=None,
                        reg_strength=0.0):
    """
    Fit a polynomial regression model with optional regularization.

    Parameters:
        degree (int): Degree of the polynomial to fit
        regularizer (str or None): 'l1', 'l2', or None
        reg_strength (float): Regularization coefficient (alpha)

    Returns:
        dict with train, val, and test MSEs, and learned
           coefficients
    """
```

Perform the following tasks.

- For three setups - no regularization, L1 and L2 regularization, repeat the below steps:
  - Fit polynomial regression models across degrees 1 to 6.
  - Plot polynomial degree vs MSE (on train and validation sets). Describe the trend you observe as degree increases.
  - For each degree, use val MSE to choose the best regularization strength.
  - Plot regularization strength (log scale) vs val MSE for best degree.
- Comment on performance improvement (if any) from regularization. Which overall experimental setup (degree, regularizer) yielded the best test performance?
- For the best setup using L1 regularization, which features had non-zero weights? List the most important predictors for GPA. Repeat the same with L2 regularization. Comment on the differences.
- You are allowed to use `sklearn.linear_model.LinearRegression`